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Optimisation model for multi-item multi-echelon supply chains with nested multi-level products

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ABSTRACT

This study proposes a mathematical optimisation model for the horizontal integration of the basic processes of production, inventory, and transportation planning on a tactical level. The model enables planning for cutting, assembly, transportation, and inventory while considering process times. Several items (raw materials, intermediate goods, finished products, and transport vehicles) are routed through a network. The general structure is defined by the layout of the supply network and the configurable multi-level bills of materials of products and transportation equipment, which can be converging, diverging, or mixed. To stress the practical applications of this study, we demonstrate how this generic model can be used and customised to solve an actual planning task in a multi-stage automotive production network using real industry data.

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1. Introduction

Supply chain planning decisions are usually made sequentially in a hierarchical order (Meyr, 2004). Planning tasks are performed separately with individual objectives, with little or no consideration given to adjacent process restrictions or global interrelations and objectives. The set of individual optimal actions often prevents the achievement of a global optimum for the planning tasks in the supply chain. The collaboration between supply chain entities, such as suppliers and customers, and company departments is often weak. Planning is based on fixed agreements (e.g. static contracts) instead of a consideration of current conditions and real flexibilities (Arshinder, Kanda, & Deshmukh, 2011).

The automotive industry is a crucial sector contributing 4% to the gross domestic product of the European Union and providing jobs for approximately 12 million people (European Commission, 2019). Thus, the automotive industry serves as an example of, and reference for, the model we propose in this study. In the automotive industry, the production quantities of final assembly lines are

defined through a central master planning process based on rough sales plans and are assigned to production sites (Meyr, 2004). Objectives include shortening customer lead times and minimising inventories of finished cars, as well as human resource goals, such as reducing overtime hours and changing shift models. In the next step, called 'order promising' or 'order scheduling', the resulting production quantities are filled with real orders. These planning objectives can differ from, and even conflict with, the previous planning objectives. Common goals include short delivery times for individually configurable options (e.g. cars with specific extras), high delivery reliability for individual orders, and generally low costs. At a fixed time before the final assembly, orders are transferred to local production sites, where their individual order sequences are defined. These local plans are used for numerous subsequent plans, such as human resource planning and short-term logistics planning, which may also have different or conflicting objectives. The initial planning stage focuses on the original equipment manufacturers' (OEM) final assembly lines. Plans that are infeasible (e.g. due to violated assembly constraints) must be recalculated in inefficient and time-consuming iteration loops, or restrictions must be considered upfront (Boysen, Scholl, & Wopperer, 2012). In contrast to the sequential-hierarchical planning process, integrated planning approaches consider two or more planning de-

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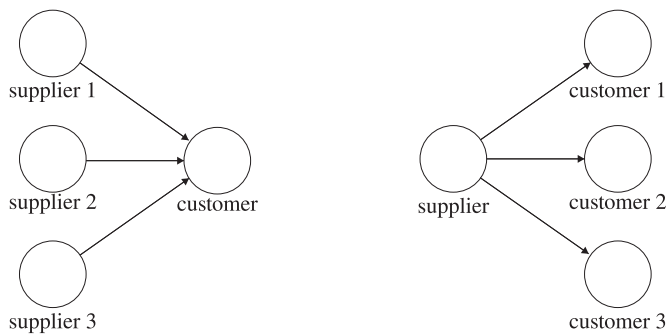


Fig. 1. Examples for converging (left) and diverging (right) structures.

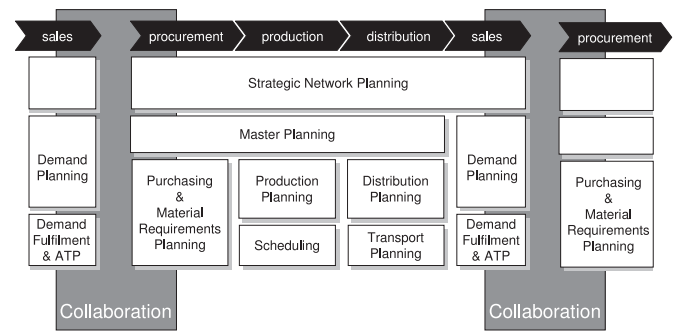


Fig. 2. Collaboration between supply chain planning matrices, see [Meyr, Wagner, and Rohde \(2008\)](#).

cisions (e.g. master planning, order scheduling, sequencing, human resource planning) or multiple process steps along the material flow (e.g. external suppliers, body shops, engine plants, assembly lines, distribution, customisation) simultaneously. This has two advantages. First, an integrated planning approach can be more efficient, because multiple constraints are taken into account directly; thus, fewer plans have to be generated in fewer iteration loops. Second, this approach better fulfills global objectives because integrated flexibilities widen the solution space and objectives that are only locally beneficial can be set aside ([Absi, Archetti, Dautère-Pérès, Feillet, & Speranza, 2018](#); [Collette & Siatry, 2004](#)). One important aspect of supply chain planning is the routing of raw materials, sub-products, and final products through supply networks, consisting of production, transportation, and inventories. Products and supply network structures can both have complex linear, divergent, convergent, or mixed (nested) structures ([Stadtler, Kilger, & Meyr, 2015a](#)). Examples of converging and diverging networks are shown in [Fig. 1](#). A converging structure consists of multiple inputs and fewer outputs. A diverging structure consists of multiple outputs and fewer inputs.

In the automotive industry, increasingly complex products with nested structures are routed through increasingly complex nested supply chains. [Serdarasan \(2013\)](#) claims that the number/variety of products, processes, and suppliers, as well as the changing needs of customers, are important drivers of complexity in supply chains. This trend is observed by [Timmer, Dietzenbacher, Los, Stehrer, and de Vries \(2015\)](#) as well, who describe international fragmentation in supply chains. The share of foreign value added in the German automotive industry grew by 13 percentage points between 1995 and 2011, from 21% to 34%. German car manufacturer AUDI AG offered 20 different models in 2005. This number had increased to 52 models by 2015. The complexity of the firm's products can be illustrated by pointing out that the 2012 model AUDI A3 could theoretically be ordered in $1.1 \cdot 10^{38}$ variants, with up to 5,000 parts assembled at the OEM's manufacturing site. The complexity of the supply chain has also increased. In 2005, a single AUDI AG manufacturing site (Ingolstadt) was supplied by factories at 1,059 different locations. This figure almost doubled over the next 10 years. The total number of parts being assembled at Ingolstadt also increased, from 9,698 in 2005 to 33,289 in 2015, while the number of containers handled daily rose from 25,719 to 71,754.

The scientific community has proposed numerous models to integrate single and multiple planning decisions or multiple process steps with complex products and supply networks. In [Section 2](#), we review the related literature. To the best of our knowledge, no general mathematical model has been developed that allows the modular horizontal integration of the most common process steps or the planning of realistic multi-step or multi-echelon production networks with complex converging, diverging, or mixed supply network structures and bills of materials (BOMs). In [Section 3](#), we

introduce a generic mathematical mixed-integer linear programming model that can be used to integrate the described planning decisions and process steps. In [Section 4](#), we demonstrate how this generic model can be customised to solve a real case in the automotive industry using realistic industry data. Conclusions and suggestions for future research directions are given in [Section 5](#). [Appendix Table 9](#) shows how to use generic model to plan basic elements of an integrated supply chain.

The main contribution to the scientific corpus is an extendable generic optimisation model. In its most simple form, the model facilitates the routing of materials with complex BOMs and transportation equipment through complex supply chains. Supply chains can be modelled using standard building blocks of the types, *production*, *inventory*, and *transportation*. The model allows the integration of different planning tasks, such as master planning, capacity planning, or scheduling, along the supply chain. Using real data, we demonstrate how the model can be extended to plan for a specific industry scenario.

2. Literature review

In this section, we discuss planning stage integration and ways of modelling products and manufacturing processes with different structures. Furthermore, we discuss the gaps in the existing research, as well as recent research proposals made by scholars in relevant fields.

[Porter \(1985\)](#) introduces the *value chain model* with horizontal linkages within a company and vertical linkages between companies, suppliers, and customers. [Rohde, Meyr, and Wagner \(2000\)](#) propose the *supply chain planning matrix*, which defines the planning tasks that need to be performed by every partner (entity) of a supply chain based on a horizontal and a vertical axis. According to [Meyr and Stadtler \(2015\)](#), the horizontal direction is defined by 'two adjacent entities within the supply chain' (p. 60). The horizontal axis of an entity can be further broken down into the functional process steps of *procurement*, *production*, *distribution*, and *sales*. The vertical direction defines the planning hierarchy, which consists of *strategic* (long-term), *tactical* (mid-term), and *operational* (short-term) planning tasks.¹ [Fig. 2](#) shows an example of the planning tasks of three adjacent entities.

Raw materials being processed in supply chains 'change in their sizes and shapes' ([Stadtler, Kilger, & Meyr, 2015b](#), p. 59). These product structures, given by the bill-of-materials (BOM), can be

¹ Some of the definitions of horizontal and vertical axes are controversial. Several authors define the horizontal axis as the number of tiers and the vertical axis as the number of entities per tier.

serial, convergent, divergent, or mixed.² Convergent structures are used in the 'high-tech, car manufacturing and white goods industry' (Atan, Ahmadi, Stegehuis, de Kok, & Adan, 2017, p. 866), whereas divergent structures are predominantly used in the consumer goods, pharmaceuticals, and process industries. According to the survey of Copil, Wörbelauer, Meyr, and Tempelmeier (2017), most research on divergent BOMs was done in the context of lot sizing. Atan et al. (2017) mention that only a few publications are applicable to multi-item, multi-echelon systems. Only a small number of papers consider divergent BOMs in generic models, which can be used for both the planning of multi-echelon inventories and multi-stage machines or work stations (Martel, 2005; Paquet, Martel, & Desaulniers, 2004; Paquet, Martel, & Montreuil, 2008; Rizk, Martel, & D'Amours, 2006; Rizk, Martel, & Amours, 2004). The simultaneous routing of products and the necessary transportation equipment, such as vehicles or containers, is a well-studied issue. To integrate the planning of material flow and transportation, the so-called *production routing problem*, four decisions must be made: '(1) how many products to manufacture each day; (2) when to visit each customer; (3) how much to deliver to a customer during a visit; and (4) which delivery route to use' (Díaz-Madroño, Peidro, & Mula, 2015, p. 518; also see Bard & Nananukul, 2010).

The basic concepts of an integrated supply chain were defined in the 1950s by Clark (1958), who extends the *inventory problem* and the *economic order quantity problem* developed by Harris (1913) into the *single-item multi-echelon inventory problem* (Arrow, Karlin, & Scarf, 1958; Bellman, 1954; Clark & Scarf, 1960; Johnson, 1954). Closely related is the task of *scheduling*, which was first described by Gantt (1918). Pinedo (2016) gives a detailed overview of recent models in his review. Many extensions and different combinations of the abovementioned models have been proposed. Some focus on specific problems, while others develop efficient solution approaches, or focus on the integration of multiple planning tasks (Calvete, Galé, & Polo, 2016; Maleki & Cruz-Machado, 2013). Popular integration approaches reviewed in the literature include the following:³

- Strategic Network Planning (Vidal & Goetschalckx, 1997)
- Process Planning and Scheduling (Khoshnevis & Chen, 1991)
- Production Planning and Distribution Planning / Vehicle Routing Planning (Chen, 2004; Chen, 2010; Díaz-Madroño et al., 2015; Gansterer & Hartl, 2018; Goetschalckx, Vidal, & Dogan, 2002; Mula, Peidro, Díaz-Madroño, & Vicens, 2010; Reimann, Tavares Neto, & Bogendorfer, 2014)
- Production Planning and Scheduling (Harjunkoski et al., 2014; Maravelias & Sung, 2009)
- Master Planning, Production Planning, and Scheduling (Dörmer, Günther, & Gujjula, 2015)
- Scheduling and Transport Planning (Ehm & Freitag, 2016; Moons, Ramaekers, Caris, & Arda, 2017)

Michael Porter claimed the following in an interview: 'We have been in the fragmentation era for the last 10 years. Hopefully sometime soon, we will get into an integration era'. (Stonehouse & Snowdon, 2007, p. 266). For future research directions, Mula et al. (2010) propose investigating the 'integration of the suppliers' nodes into the supply chain's optimization models' (p. 386) and the modelling of different transportation forms, such as milk runs or grouping. They observe that the models have rarely been

applied to real cases. Reimann et al. (2014) propose the integration of 'multi-machine and multi-product problems or combinations of tactical and operational problems' (p. 210). Díaz-Madroño et al. (2015) propose the synchronisation of tactical and operational models, including inventories and warehouses, and the 'consideration of vehicle routing planning for inbound logistic processes' (p. 533), especially for 'those industries with complex bills of materials, such as automobile... firms' (p. 533). They also propose the usage of different transport modes, such as long-distance transport processes or milk runs and cross docking. Atan et al. (2017) propose the development of 'policies for multi-echelon systems' consisting of 'multiple echelons upstream of the original equipment manufacturer assembling the end product' (p. 877).

To the best of our knowledge, no mathematical models have been proposed to route products, sub-products, and raw materials with nested multi-level BOMs and transportation equipment through divergent and convergent multi-echelon supply networks. The main contribution of this study is a generic optimisation model, which can be used as a reference model for routing these complex products through complex supply chains. The model is provided in a basic generic form. It facilitates the planning of material flow through supply chains, which comprises any combination of the basic building blocks, as described in Table 9. Furthermore, the model is designed such that it can be extended to integrate other planning tasks, as presented in the supply chain planning matrix in Fig. 2. The motivation for possible extensions to the basic model can originate from two sources: in response to industrial requirements (as demonstrated in Section 4) and as required by the scientific community to integrate further planning tasks.

3. Model description

This study should help fill the abovementioned research gaps by developing a model with the following requirements:

- Items (e.g. raw materials, intermediate goods, finished products) and transportation equipment (e.g. containers, trucks, ships) are routed through a network.
- The network can be a serial, convergent, divergent, or mixed/nested arrangement of basic process steps/elements.⁴ The basic elements can be used in any logical combination. This enables the modelling of complex supply networks.
- The structures of the routed items are defined by multi-level variant BOMs. These can be serial, convergent, divergent or mixed/nested.
- Items can be defined individually (e.g. build-to-order [BTO] cars) or as grouped orders with identical options (e.g. batches of build-to-stock [BTS] cars).
- Capacities are considered (e.g. for master production planning or capacity planning of suppliers or machines)⁵.
- Lead times are considered (e.g. for manufacturing times, transportation times).
- Items can be stored in inventories.
- The model is generic: It must be extendable towards long-term strategic planning (e.g. strategic network planning, evaluation of long-term sales plans, line-balancing of flow shops) and short-term operational planning (e.g. sequencing of flow shops, short-term materials disposition).

² In other publications a mixed product structure is called *nested*, *general* or *acyclic*. In scheduling theory, the precedence constraints are called *chains*, *intree* and *outtree* (e.g. Pinedo, 2016).

³ To allow comparability, we classify the papers based on the notation of the supply chain planning matrix wherever possible, as shown in Fig. 2.

⁴ For details of these elements and examples of how to model them using the mathematical formulation see Table 9.

⁵ Possible ways of expanding the model to facilitate the planning of additional tasks are shown in Section 5.

The finite planning horizon is an ordered list T of $t = 1, 2, \dots$ equal time periods. A time period t can be defined as a point at the beginning, the middle or the end of a planning period. For simplicity, we define t as the beginning of a planning period in the remainder of this work. This means inventories are defined at the beginning of period t , whereas processes are defined as being performed during period t . The granularity of periods t should be chosen such that it accurately fulfils the underlying planning process. Minutes, shifts, days, weeks, or other intervals could be chosen.

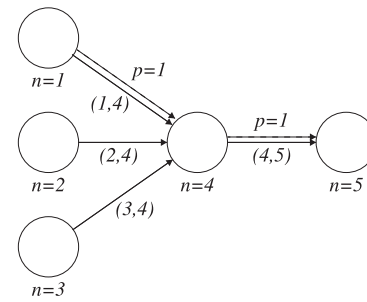
In the basic model formulation, items are routed through a network. The items are indicated by the set P , where each item p represents a specific product (e.g. $p = \text{wheel}$) or certain transportation equipment (e.g. $p = \text{truck}$). The network is defined by a directed graph $G = (N, A)$, with nodes $n \in N$ and edges $A \subseteq \{(i, j): i, j \in N, i \neq j\}$, where i is the starting node and j is the ending node. An item p can be stored at node n over time period t . The resulting inventories are represented by the decision variable inv_{npt} (inventory of item p at node n at the beginning of period t). The initial inventory of item p at node n at the beginning of period $t = 0$ is given by inv_{np} . Edges have two characteristics:

- An item p can flow through edges (i, j) . The flow is calculated as the decision variable f_{ijpt} , which defines the amount of the item p that starts flowing on edge (i, j) during period t . The process lead time (or throughput time) l_{ij} defines the time periods t that elapse between the item's entry to the edge at node i to its exit of the edge at node j .
- The capacity constraint c_{ijt} limits the amount of the item p that starts flowing through the edge (i, j) at period t .

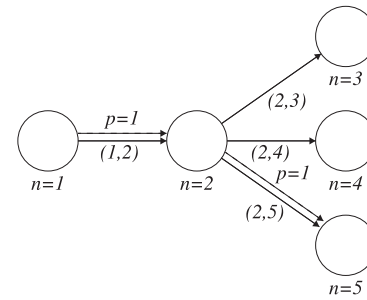
The model minimises the total inventory inv_{npt} and flow of materials f_{ijpt} . Both variables directly influence the total costs of the model's planning results: Low inventories inv_{npt} result in low inventory holding costs. Minimising the flow of materials f_{ijpt} means low manufacturing and transportation costs. Factors and weights are neglected to improve the model's readability.

The structure of the supply network can be converging, diverging, or mixed/nested, as shown in Fig. 3. Fig. 3a shows a network with a converging structure. The dashed line shows the flow of a particular item $p = 1$. It flows from node 1 on edge $(1,4)$ towards node 4 and further down towards node 5 on edge $(4,5)$. Fig. 3b shows a similar example with a diverging structure. Item 1 flows from node 1 via node 2 towards node 5.

Each item p must be assigned to a superior item group $g \in G$. The parameter g_p assigns each product p to a specific group g . For instance, the items *wheel_small* and *wheel_big* can be assigned to the group *wheel*.⁶ In Fig. 4, the three item groups *VIN*,⁷ *wheel*, and *body* are represented by the outer boxes. Assigned items p are illustrated by the smaller inner boxes. It is also possible to have only a single item p within a group g (as shown for the group *body*). Transportation equipment is a special case. All items indicating transportation equipment (e.g. $p = \text{truck}$) must be assigned to the group $g = \text{transport}$. The relation between multiple groups g can also be serial, diverging (e.g. cutting operations), converging (e.g. assembling operations), or mixed (e.g. stamping operations on a transfer press, with coils and cutting fluid as inputs and left and right parts as outputs). This structure is defined by multi-level BOMs. They define which parent (or mother) groups y consist of which child groups x . A multi-level BOM combines multiple



(a) Converging



(b) Diverging

Fig. 3. Structure of networks.

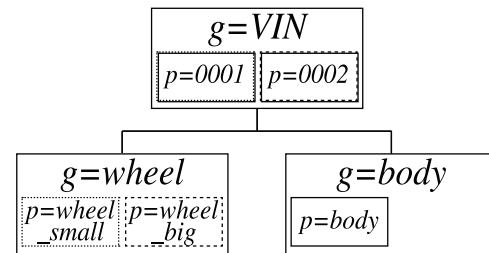


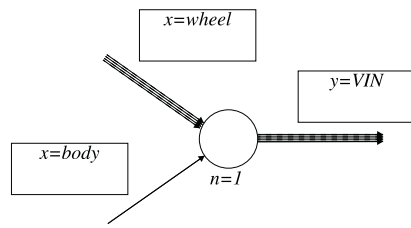
Fig. 4. Example of multi-level BOMs: Items (bottom inner boxes) of item groups (bottom outer boxes) are assembled to mother items (top inner boxes).

levels of mother-child relations within a single list. The example in Fig. 4 shows a two-level BOM. The items of group *VIN* consist of child items (or sub-items) of the groups *wheel* and *body*. This means that one (or more) specific item of the child group *wheel* (e.g. $p = \text{wheel_small}$) and one specific item of the child group *body* are needed simultaneously to assemble one specific item of the mother group *VIN* (e.g. $p = 0001$). If we were to include another level, such as the groups *tyre* and *rim*, as sub-groups of *wheel*, we would yield a multi-level BOM.

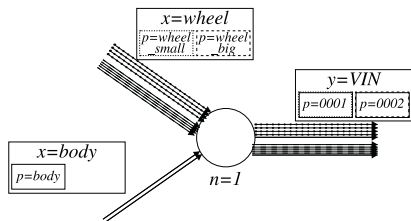
To define a location in the network where an operation (e.g. cutting or assembling) is conducted, mother-child relations are assigned to at least one particular node n . At this particular node n , we define the 'input group' (child group) x and the 'output group' (mother group) y . We call this assignment an 'operation' o and define it as a tuple $o = (n, x, y)$. If the exact type and quantity of items chosen from a group vary from product to product, they are defined individually by variant BOMs. For example, the car $p = 0001$ uses the item $p = \text{wheel_small}$, whereas car $p = 0002$ uses the item $p = \text{wheel_big}$ (see Fig. 4). This allows the handling of products with different sub-items. Individual child items c are assigned to corresponding mother items m by setting $B \subseteq \{(m, c): m, c \in P, m \neq c\}$. To indicate that individual child items are used, the

⁶ This concept is motivated by the product definition in the automotive industry. At the Volkswagen AG a car is completely defined by choosing exact items (so called codes) for about 160 item groups.

⁷ VIN is the Vehicle Identification Number, which represents a specific finished car.



(a) Two operations, being performed simultaneously at node $n = 1$.



(b) Particular items p being processed by two operations on node $n = 1$.

Fig. 5. Examples of how to integrate the item structure, defined by BOMs into the network structure.

Table 1
Operations O at nonde $n = 1$.

Field	Value
input group x	wheel
output group y	VIN
input quantity in_{nxy}	4
output quantity out_{nxy}	1
binary parameter β_{nxy}	1
input group x	body
output group y	VIN
input quantity in_{nxy}	1
output quantity out_{nxy}	1
binary parameter β_{nxy}	0

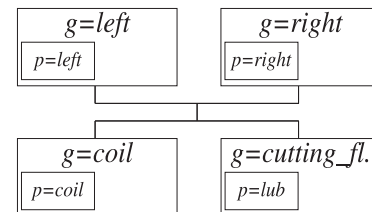
operation o must be flagged with the binary parameter $\beta_{nxy} = 1$. Individual input quantities are defined by q_{mc} .

Fig. 5 a shows an example of two operations performed simultaneously at node $n = 1$. Each group is represented by a box, whereas the flow is represented by the dashed lines. The first operation (on the top) uses the group *wheel* as input group x and *VIN* as output group y . The second operation (on the bottom) uses the group *body* as input group x and *VIN* as output group y . This operation is represented by the straight lines. Multiple arrows are plotted to illustrate the quantities of each item required to perform the operation. In this example, we define the two operations, 'body assembly to VIN' (1, *wheel*, *VIN*) and 'wheel assembly to VIN' (1, *body*, *VIN*), to model the single physical operation 'body and wheels assembled to VIN'. Both operations must be defined individually, although they might be conducted simultaneously. The data for both operations are shown in Table 1.

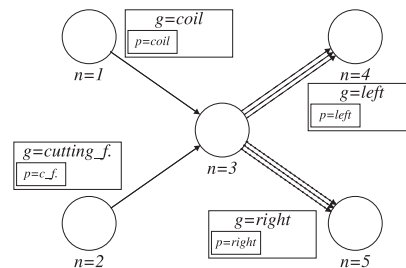
The groups and the quantities of this example can be seen as a placeholder for the actual items p , that flow through the network. For each operation, values in_{nxy} and out_{nxy} must be defined. They represent the quantity of items needed as input and the quantity of output items. Imagine two customers buy cars 0001 and 0002 of the group *VIN*. They can choose between two quantities of the items of the group *wheel*. Car 0001 is ordered with four small wheels, whereas car 0002 is ordered with five big wheels (including one spare wheel). To indicate, that a specific number of particular items p is chosen for the 'wheel assembly to VIN' operation,

Table 2
Variant bills-of-materials (BOM) for two mother items m .

Field	Value
mother item m	0001
child item c	wheel_small
ind. input quantity q_{mc}	4
mother item m	0002
child item c	wheel_big
ind. input quantity q_{mc}	5



(a) Nested BOM structure with two mother items and two child items



(b) Parallel assignment of all operations to node $n = 3$ with identical $\alpha_{n=3,y}$

Fig. 6. Example of nested item structure, where multiple items are produced by a single operation.

the corresponding binary parameter $\beta_{1,wheel,VIN}$ is set to one. For this flagged operation, the individual input quantities (q_{mc}) for all possible items p (the customer orders 0001 and 0002) must be set. They now overrule the general input quantity (in_{nxy}) of that operation. This example is shown in Fig. 5b. The data for the variant BOMs are shown in Table 2. To summarise, input quantities and items can be defined in two ways:

- They can be defined generally by the output quantity for an operation at a specific node n by out_{nxy} .
- Or they are defined specifically for an item by its variant BOM, by defining the individual input quantity q_{mc} and setting the parameter β_{nxy} to one.

Some processes require multiple items p to flow simultaneously out of a node, e.g. left and right parts are always produced simultaneously from a single coil and some additive material as input (see Fig. 6a). To guarantee the simultaneous flow of multiple output item groups, we introduce the parameter α_{ny} . When α_{ny} is set to equal values for multiple output groups y , their operations are always forced to be performed simultaneously. An example is shown in Fig. 6b. One item of the group *coil* and one item of the group *cutting_fluid* are used to simultaneously produce three items of the group *left* and three items of the group *right*. The parameter α_{ny} must be set to the same value for all output groups y . This forces the items *right* and *left* to flow out of the node simultaneously whenever at least one of the two items is pulled.

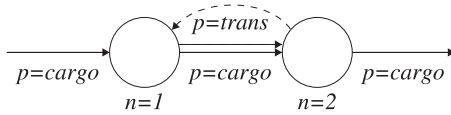


Fig. 7. Transportation example.

To plan the flow of transportation equipment (such as trains, trucks, ships, and containers), the items of group *transport* are routed through the network the same way as other items. Possible routes must be defined by operations *o*. Every item *p* of the type transportation equipment (with *p* in the group $g_p = \text{transport}$) has a capacity of s_p . Every item *p* in $g_p \neq \text{transport}$ that needs to be transported has a demand on this capacity $s_p > 0$. If an item does not need to be assigned as transportation equipment, the demand is set to zero $s_p = 0$. Fig. 7 shows an example of an item $p = \text{cargo}$ of group $g_{\text{cargo}} = \text{transport}$ that needs to flow from node 1 via the edge (1,2) to node 2 (straight line). It consumes a transportation capacity of $s_{\text{cargo}} = 1$. Transportation equipment $p = \text{trans}$ can transport items from node 1 to 2 and back to node 1 (dashed lines). Note that resulting plans depend on the initial location of the transportation equipment (given by the initial inventory parameter $inv_{n,p=\text{trans}}$) and the process lead time of the edges given by $l_{1,2}$ and $l_{2,1}$.

All the characteristics mentioned above can be combined. The formulations for some basic supply chain elements are presented in Table 9. The entity relationship model of the data is shown in Appendix Fig. 12. The mathematical model is defined as follows.⁸

$$\min \sum_{t \in T} \sum_{p \in P} \left(\sum_{n \in N} inv_{npt} + \sum_{(i,j) \in A} f_{ijpt} \right) \quad (1)$$

subject to :

$$\sum_{p \in P} f_{ijpt} \leq c_{ijt} \quad \forall (i,j) \in A, t \in T \quad (2)$$

$$inv_{np,0} = inv_{np} \quad \forall n \in N, p \in P \quad (3)$$

$$\begin{aligned} inv_{npt} = & inv_{np,t-1} - d_{npt} + \sum_{\substack{(i,n,p) \in F: \\ l_{in} < t}} f_{inp,t-l_{in}} \\ & - \sum_{\substack{(n,g_p,y) \in O: \\ \beta_{ngpy}=0}} \sum_{\substack{(n,j,p') \in F: \\ g_{p'}=y}} \frac{in_{ngpy}}{out_{ngpy}} \frac{f_{njpt}}{\sum_{\substack{(n,g_p,y') \in O: \\ \alpha_{ny'}=\alpha_{ny}}} 1} \\ & - \sum_{\substack{(n,g_p,y) \in O: \\ \beta_{ngpy}=1}} \sum_{\substack{(n,j,p') \in F: \\ g_{p'}=y}} \sum_{(p',p) \in B} \frac{q_{p'p}}{out_{ngpy}} \frac{f_{njpt}}{\sum_{\substack{(n,g_p,y') \in O: \\ \alpha_{ny'}=\alpha_{ny}}} 1} \end{aligned} \quad \forall n \in N, p \in P, t \in T \quad (4)$$

$$\frac{f_{ijpt}}{\sum_{(i,x,g_p) \in O} out_{ixg_p}} = \frac{f_{ij'p't}}{\sum_{(i,x,g_{p'}) \in O} out_{ixg_{p'}}} \quad \forall (i,j,p) \in F, \forall (i,j',p') \in F : \alpha_{ig_p} = \alpha_{ig_{p'}}, \forall t \in T \quad (5)$$

Set of indices

T	Ordered list of time periods $t = 1, 2, \dots$
P	Set of items $p \in P$
G	Set of item groups $g \in G$
N	Set of nodes $n \in N$
A	Set of tuples for edges $A \subseteq \{(i,j) : i,j \in N, i \neq j\}$ with starting node <i>i</i> and ending node <i>j</i>
B	Set of tuples for variant BOMs $B \subseteq \{(m,c) : m,c \in P, m \neq c\}$ with mother item <i>m</i> and child item <i>c</i>
O	Set of tuples for operations $O \subseteq \{(n,x,y) : n \in N, x,y \in G\}$ with input item group <i>x</i> and output item group <i>y</i> at node <i>n</i>
F	Set of tuples for possible flow of material $F \subseteq \{(i,j,p) : (i,j) \in A, p \in P\}$ of each item <i>p</i> per edge (i,j)

Parameters

l_{ij}	Process lead time at edge (i,j)
d_{npt}	Demand of item <i>p</i> at node <i>n</i> at period <i>t</i>
inv_{np}	Initial inventory of item <i>p</i> at node <i>n</i>
c_{ijt}	Capacity at edge (i,j) at period <i>t</i>
g_p	Group g_p of item <i>p</i>
s_p	Exception: $g_p = \text{transport}$ means transportation equipment with the transportation capacity s_p Transportation size (demand) for each item <i>p</i> Exception: For transportation equipment ($g_p = \text{transport}$) s_p represents the transport capacity
in_{nxy}	Input quantity at node <i>n</i> for input group <i>x</i> and output group <i>y</i>
out_{nxy}	Output quantity at node <i>n</i> for input group <i>x</i> and output group <i>y</i>
q_{mc}	Individual input quantity defined by configurable multi-level BOMs <i>B</i> : Defines how many child items <i>c</i> (input) to be used per mother item <i>m</i> (output) (β_{nxy} must be set to 1 to use the individual input quantity)
α_{ny}	Parameter to force multiple output item groups <i>y</i> to flow simultaneously out of a node <i>n</i> , e.g. to model cutting operations: All the operations of any output item group <i>y</i> with the same α are forced to flow simultaneously at a node <i>n</i>
β_{nxy}	Binary parameter to force the operation (n,x,y) to use individual child items <i>c</i> as input group <i>x</i> , mother items <i>m</i> as output group <i>y</i> and / or assembly quantities q_{mc} as the number of child items to be assembled, for instance to model options (defined by BOMs <i>B</i>) 0: Use quantities of item groups (in_{nxy}, out_{nxy}) 1: Use optional assembly quantities (q_{mc}, out_{nxy})

Decision variables

f_{ijpt}	Flow of item <i>p</i> at edge (i,j) during period <i>t</i>
inv_{npt}	Inventory of item <i>p</i> at node <i>n</i> at the beginning of period <i>t</i>

⁸ In a slight abuse of notation, we use $(i,n,p) \in F$ for a given *n* as a shorthand for $(i,j,p) \in F : j = n$.

$$\sum_{\substack{(i,j,p) \in F: \\ g_p = \text{transport}}} \sum_{p \in P} f_{ijpt} \cdot s_p \geq \sum_{\substack{(i,j,p) \in F: \\ g_p \neq \text{transport}}} \sum_{p \in P} f_{ijpt} \cdot s_p \quad \forall (i, j) \in A, t \in T \quad (6)$$

$$f_{ijpt} \in \mathbb{N}_0 \quad \forall (i, j) \in A, p \in P, t \in T \quad (7)$$

$$\text{inv}_{npt} \in \mathbb{N}_0 \quad \forall n \in N, p \in P, t \in T \quad (8)$$

The objective function (1) minimises the inventory inv_{npt} over all time periods $t \in T$, items $p \in P$, and nodes $n \in N$, as well as the flow of material f_{ijpt} over all time periods $t \in T$, items $p \in P$, and edges $(i, j) \in A$. Constraint (2) ensures that the flow of material f_{ijpt} over all items p starting in period t from node i towards node j does not exceed the edge's capacity c_{ijt} . Constraint (3) defines the initial inventories $\text{inv}_{np,t=0}$. Constraint (4) represents the inventory flow balance equation for each node $n \in N$, item $p \in P$, and period $t \in T$. The left side of the equation represents the inventories inv_{npt} at all nodes n for all items p at any given time period t . They consist of the following five elements (right side of the equation):

- The inventories $\text{inv}_{np,t-1}$ of the previous period $t - 1$.
- The (negative) external demand d_{npt} .
- The input flow of all items p that started flowing in period $t - l_{in}$ at node i and arrive at node $j = n$ in period t . The condition $l_{in} < t$ prevents input flows at non-positive time periods.
- The (negative) pulled items resulting from the output flow into the network without options (second row, with $\beta_{ngpy} = 0$).
- The (negative) pulled items resulting from the output flow into the network with options (third row, with $\beta_{ngpy} = 1$).

We will describe the second and the third rows in detail. For both rows, the first summands define which subsets of operations O , material flows F , and (individual) BOMs B are used to map the outflows $f_{ijp't}$ to the inventories inv_{npt} . The first multiplier defines the ratio per outflow, which is pulled out of the inventory n . The first summand of both rows selects the operations $(n, g_p, y) \in O$ that satisfy the following criteria:

- n : The node of the operation selected must be equal to the node of the inventory n .
- $x = g_p$: Only the input item group x that equals the group g_p of the item p in the inventory is considered. This means that only the items p of the group g_p are pulled out of the inventory.
- All remaining output item groups y are selected.
- $\beta_{ngpy} = 0$: If the parameter is set to $\beta_{ngpy} = 0$, no individual child items or quantities are used. Hence, the second row is used.
- $\beta_{ngpy} = 1$: If the parameter is set to $\beta_{ngpy} = 1$, an operation uses individual child items c as inputs or individual quantities of child items q_{mc} . Hence, the third row is used.

Based on the selected tuples of operations (n, g_p, y) , the indices $F \subseteq \{(n, j, p'): j \in N\}$ of the material outflow $f_{njp't}$ are selected. The quotes indicate that a different set of products p' is used, as defined by the outer for all statement ($\forall p \in P$).

- $i = n$: The model sums up over all flows whose starting nodes i equal the inventory node n .
- $g_{p'} = y$: Only those flows are considered, whose item groups g_p are subsets of the selected operations outflow y .
- All flows with the remaining items p' flowing towards any ending node j are considered. These need not be equal to the items p , which are pulled out of the inventory n .

In the third row, the tuples with individual entries in the BOMs $(m, c) \in B$ are selected based on the following criteria (in case $\beta_{ngpy} = 1$):

- $m = p'$: The mother item m must be equal to the item of the material outflow p' .
- $c = p$: The child item c must be equal to the item of inventory p .

These criteria guarantee that the tuples of operations $(n, g_p, y) \in O$ of material flow $(n, j, p') \in F$ and (optionally) of BOMs $(p', p) \in B$ are selected for assigning the outflows $f_{njp't}$ to the correct items p , which are pulled out of the inventory inv_{npt} . The actual quantities pulled out of the inventory are calculated by multiplying the outflows $f_{njp't}$ with the following factors:

- $\frac{\text{in}_{ngpy}}{\text{out}_{ngpy}}$: For operations where no individual BOM is used ($\beta_{ngpy} = 0$), outflows $f_{njp't}$ are multiplied by the necessary input quantity of operation in_{ngpy} and divided by the output quantity of the operation out_{ngpy} .
- $\frac{q_{p'p}}{\text{out}_{ngpy}}$: For operations where individual BOM variants are used ($\beta_{ngpy} = 1$), outflows $f_{njp't}$ are multiplied by individual input quantity $q_{p'p}$ and divided by the output quantity of the operation out_{ngpy} .

If multiple output item groups y are forced to flow simultaneously out of node n (by setting the parameters α_{ny} identically), the assigned input items p are pulled only once out of the inventory n . This is guaranteed by dividing the outflows $f_{njp't}$ by the number of simultaneous outflows. This number is calculated by counting the number of tuples $(n, x, \alpha) \in O$ that satisfy the following conditions:

- n : The node of the operations equals the node of the inventory n .
- $x = g_p$: The input item group x equals the item group, which g_p is pulled out of the inventory. The index x is already defined by the first summand.
- $\alpha_{ny'} = \alpha_{ny}$: The parameter $\alpha_{ny'}$ equals the parameter $\alpha_{ny'}$ of the output item group y of the first summand. This ensures that the flow $f_{njp't}$ is only divided by the number of simultaneous outflows with the same α_{ny} .

Constraint (5) guarantees that all outflows, with an identical parameter α_{ny} are flowing simultaneously via a pairwise comparison of all flow variables f_{ijpt} and $f_{ij'p't}$. All pairs with identical $\alpha_{igp} = \alpha_{igp'}$ are considered. To normalise the flows, they are divided by the related output quantities out_{ixg_p} . Constraint (6) ensures that the size of the transported items does not exceed the available capacities of the transportation equipment for all edges $(i, j) \in A$. The left side of the constraint represents the available transportation capacity, given by the flow f_{ijpt} of the transportation equipment ($g_p = \text{transport}$) times their capacities s_p . The demand for transportation size is given by the flow of items $f_{ijp't}$ with $g_p \neq \text{transport}$ multiplied by their transportation sizes s_p . Constraints (7) and (8) ensure that the values of the decision variables are all non-negative integers.

Physical supply chains can serve as arrangements of *production*, *inventory*, and *transportation* elements (Arnold, Isermann, Kuhn, Tempelmeier, & Furmans, 2008). A significant advantage of the generic formulation is that it allows the horizontal integration of these elements and planning in a single process step. Basic building blocks are shown in Appendix Table 9.

4. Production scheduling: A case study within a multi-echelon automotive production network at Volkswagen AG

This section presents a case study of production scheduling in the presence of short-term supply shortages in a multi-echelon automotive production network at Volkswagen AG. The case study

aims on demonstrating how the generic model presented in the previous section can be customised for specific applications. We first put forward the motivations for the use case and outline the planning task. Subsequently, we specify the item and network structure. Then, we explain the model requirements, the application of the model, the data sets used, and the implementation. Finally, we analyse the results and conclude with a discussion of this case study.

4.1. Motivation

In January 2019, production ceased in several plants of Volkswagen AG due to a labour strike in an Hungarian engine plant (Handelsblatt, 2019). This example shows how short-term events within a multiple-stage production network can adversely affect the stability of production schedules. Such occurrences not only affect the OEM's plant, but also trigger far-reaching chain reactions within the entire network (Hozak & Hill, 2009). This phenomenon is known as 'schedule instability' (SI), provoked by uncertainty within and outside production networks. Uncertainty rises due to increasing complexity and dynamism (Choi, Dooley, & Rungtusanatham, 2001; Duncan, 1972). In automotive production networks, the OEM and its suppliers are linked directly (see Fig. 2). In turn, suppliers in the upstream supply chain are indirectly linked by unique customer orders and the associated BOM. Thus, supply shortages do not affect only single OEM-supplier relations. Changes in the OEM's weekly production schedule affect the material requirements planning and the associated quoted call-orders of other suppliers. This means that the initiation of countermeasures may lead to cascading disturbances. So far, the short-term supply shortage planning, in practice, is unsystematic, experience-based, and personnel- and time-intensive. Due to the personnel and time intensity of this activity, the extent of the shortages and the effects of the shortage measures cannot completely be grasped, evaluated, and compared on the basis of different scenarios. This leads to sub-optimal, locally oriented and slow reactions. Thus, Moetz, Meinschmidt, and Otto (2018) emphasise the importance of a network-oriented corporate production strategy for managing SI. To enrich the debate and to demonstrate the ability of the generic model to be used in actual planning tasks, our generic framework for supply chain master planning is applied to a multiple-echelon automotive production network to solve the task of short-term supply shortage planning. The overall motivation is to predict emergent bottlenecks and resulting production downtime through a multiple-period process with transparency at all tiers.

4.2. Description of planning task

As shown in Fig. 8, the present planning task involves an excerpt of a European automotive production network that comprises one OEM, four first-tier suppliers, and two second-tier suppliers. A key characteristic of the European automotive industry is its BTO-driven production strategy. Hence, and in the case of short-term supply shortage, the satisfiability of customer orders has to be considered. The planning task in this scenario is triggered by a short-term supply shortage caused by quality deficiencies in delivered battery cells. The weekly shipment of battery cells cannot be used for further production operations. Table 3 provides details on the supply shortage scenario. Due to the interrelated BOMs, such occurrences can affect the whole production network. Following the supply chain planning matrix described above (see Fig. 2 in Section 3), the functional process step are *master planning*, *production scheduling*, *material requirements planning*, and *demand fulfilment*. The objectives of this case study are:

Table 3
Information about the supply shortage scenario.

Type	Value
Periods T	$t=61, \dots, 74$
Supply shortage cause	quality deficiencies
Supply shortage group g	$g=\text{battery_component}$
Supply shortage amount s	$s=860$
Supply shortage period t	$t=59$
First affected node n	$n=\text{battery-supplier_prod}$

Table 4
Variant BOMs for individual orders, shown as mother items m (left column) and its child items c (mid columns). Each column represents a item group g . The right column shows the period t of the promised delivery date (d_{npt}).

Mother	Child				Period
m_{car}	C_{engine}	C_{gear}	C_{seat}	$C_{bat.}$	t
00001	DK8	G0C	Q4H	-	61
:	:	:	:	:	:
01430	D60	G1D	Q4H	BEV	63
:	:	:	:	:	:
28000	DK8	G0K	Q2J	-	74

- the demonstration of the model's ability to be used for a short-term planning task in production networks,
- the evaluation of the supply shortage's impact on the production network,
- the examination of the OEM's production schedule stability, and
- the development and evaluation of possible supply shortage steering strategies.

4.3. Description of the item structure

The relevant items are derived from the customer orders and respective BOMs for the secondary requirements across all value-added stages. Overall, $|P| = 28,049$ items (e.g. 0001, *leather1*, *Q2H*) and seven different item groups (e.g. *VIN*, *seat*, *seat_component*) are considered. Each item p is assigned to one of the seven item groups (e.g. $p = D80$ is an item of the item group $g = \text{engine}$). The item structure of each item $p \in P$ is specified by a variant BOM B . For this planning scenario, a total of 28,000 different cars (with item group $g = \text{VIN}$) and 49 different assembly parts of six item groups are considered. As shown in Table 4, each customer order is indexed by an order number (p of group $g = \text{VIN}$), technically defined by its variant BOM (as shown in Table 4), and scheduled by its promised delivery date at its delivery location (demand d_{npt} during period t at node $zp8$).

The variant BOMs define the relation of the mother item $m \in P$, its child items $c \in P$, and their individual input quantities q_{mc} . The individual input quantity overrides the general input quantities in_{nxy} , when the parameter β_{nxy} is set to one. Taking the customer order of item $p_{car}=1430$ as an example, this indicates that the item $m = 01430$ belongs to item group $g = \text{VIN}$. As shown in Table 5, each mother item (e.g. *VIN* or *battery*) requires individual child items c and quantities q_{mc} . A car requires one child item c of each of the four different item groups $g \in \{\text{engine}, \text{gear}, \text{seat}, \text{battery}\}$ with $p_{engine}=D60$, $p_{gear}=G1D$, $p_{seat}=Q4H$, and $p_{battery}=BEV$. The car $p = 00001$, for example, requires one item $p = DK8$ of the item group $g = \text{engine}$, one item $p = G1D$ of the item group $g = \text{gear}$, and so on. Likewise, for the upstream supply chain, the battery $p_{battery} = BEV$ requires multiple child items, *cell1*, *cell2*, and *cell3*.

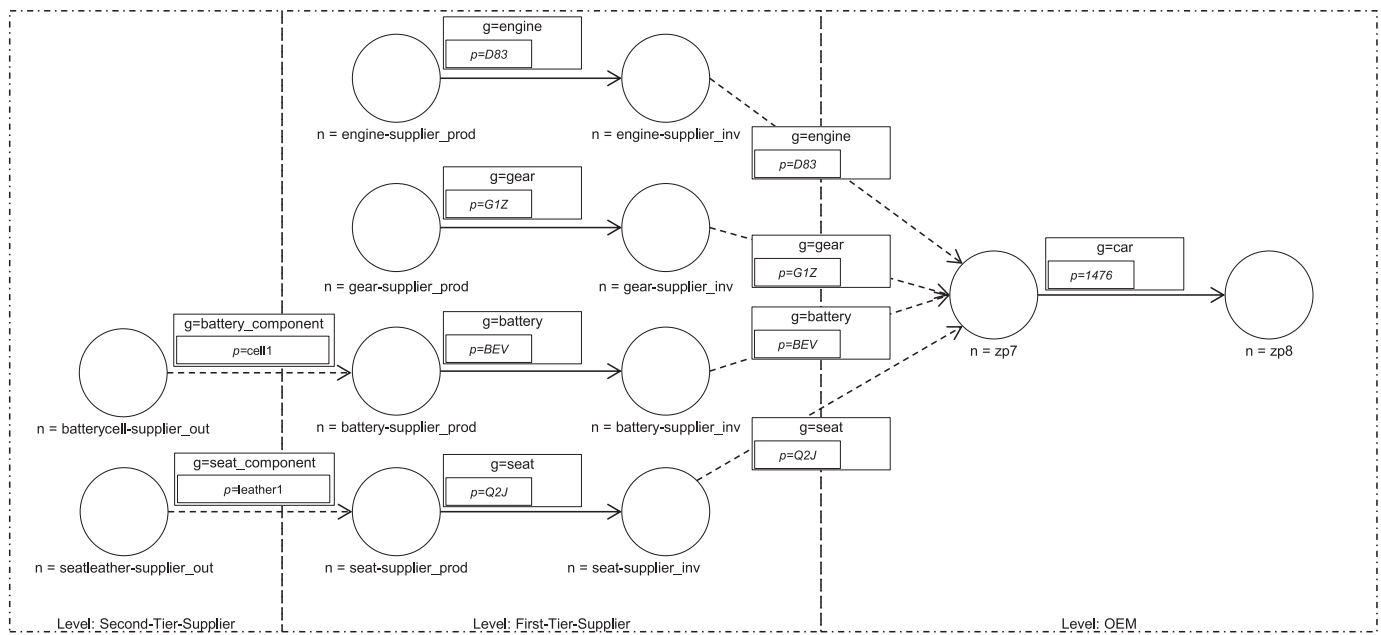


Fig. 8. Network and item structure of automotive production case study.

Table 5
Exemplary item structure (BOM).

Moth. itm m	Child itm. grp. g	Child item c	q_{mc}
01430	engine	D60	-
01430	gear	G1D	-
01430	seat	Q4H	-
01430	battery	BEV	1
BEV	batt.-compnt.-1	cell1	-
BEV	batt.-compnt.-2	cell2	-
BEV	batt.-compnt.-3	cell3	-

4.4. Description of network structure

The network has the structure of a three-echelon converging inter-organisational production network. Fig. 8 illustrates the resulting network structure. Twelve nodes N (e.g. $n=zp7$) and eleven edges A (e.g. $(i, j)=(zp7, zp8)$) are implemented. Nodes with the suffix *out* are the second-tier suppliers, while the first-tier suppliers production nodes are marked with the suffix *prod*. Inventory nodes are marked with the suffix *inv*. Each node is characterised by operations o , which specify the input groups x , the output groups y , the parameters α_{nxy} , the input quantities in_{nxy} , and the output quantities out_{nxy} . As exemplified in Table 6, operation o at node $n=battery-supplier_prod$ always requires 20 items of item group $g=battery-component$ to produce one item of item group $g=battery$. Since there are no operations o with multiple output item groups y that flow simultaneously out of a node n , α_{nxy} is set to 0 for $n \in N$. The time horizon T is divided into periods t , whereby one period t conforms to the length of one day. An edge $(i, j) \in A$ is specified by its period t -related capacity c_{ijt} and its process lead time l_{ij} . As shown in Table 7, for edge $(battery-supply_out, battery-supply_prod)$ with lead time $l_{ij}=20$, the capacity c_{ijt} in period $t=67$ is 490. An edge (i, j) subsequent to a production node with $n=i$ represents a production process, and an edge (i, j) subsequent to a transportation node with $n=i$ represents a delivery process.

Table 6
Excerpt of data defining the operations O of the network.

Field	Value
node n	zp7
input group x	engine
output group y	VIN
α_{nxy}	1
input quant. in_{nxy}	1
output quant. out_{nxy}	1
node n	zp7
input group x	BEV
output group y	VIN
α_{nxy}	1
input quant. in_{nxy}	-
output quant. out_{nxy}	1
node n	batt.-supplier_prod
input group x	batt.-compnt.-1
output group y	battery
α_{nxy}	1
input quant. in_{nxy}	10
output quant. out_{nxy}	1
node n	batt.-supplier_prod
input group x	batt.-compnt.-2
output group y	battery
α_{nxy}	1
input quant. in_{nxy}	6
output quant. out_{nxy}	1

OEM:

With up to $3.44 \cdot 10^{32}$ possible variants of the OEM's compact models, the OEM's general lot-size is one. To handle this complexity and to reduce inventories, the production is BTO. To simplify, the assembly line is modelled by edge $(i, j)=(zp7, zp8)$ with lead time $l_{ij}=0$. Node $n=zp7$ is the starting node of the assembly line with an inflow of item groups $g=engine$, $g=gear$, $g=battery$, and $g=seat$ and an outflow of item group $g=car$. The OEM produces three different compact models, one compact sport utility vehicle (SUV), and one midsize SUV. Two of the compact models have an internal combustion engine (ICE), whereas the third one is a bat-

Table 7Data for edges A exemplified by the battery supply chain at period $p = 6$.

Field	Value
starting node i	bat.-cell-supl._out
ending node j	battery-supplier_prod
process lead time l_{ij}	20
group g	battery_component
capacity at edge c_{ijt}	490
starting node i	bat.-cell-supl._prod
ending node j	bat.-cell-supl._inv
process lead time l_{ij}	1
group g	battery
capacity at edge c_{ijt}	70
starting node i	bat.-cell-supl._inv
ending node j	zp7
process lead time l_{ij}	1
group g	battery
capacity at edge c_{ijt}	70
starting node i	zp7
ending node j	zp8
process lead time l_{ij}	0
group g	car
capacity at edge c_{ijt}	2000

tery electric vehicle (BEV). Overall, the OEM produces 14,000 cars per week.

Battery supply chain:

The contemplated battery supply chain includes a second-tier battery cell supplier and a first-tier battery supplier. Table 7 illustrates the battery supply chain that starts at the second-tier supplier and ends at the OEM. The battery supplier produces batteries for the BEV. As specified by the operations at node *battery-supplier_prod*, each battery m consists of 20 battery cells c , comprising three different battery cell items p (see Table 6). The production lead time l_{ij} of edge $(i, j) = (\text{battery-supplier_prod}, \text{battery-supplier_inv})$, as well as the transportation lead time l_{ij} of edge $(i, j) = (\text{battery-supplier_prod}, \text{battery-supplier_inv})$, are 1. Since the supply of battery cells is a bottleneck within the whole battery supply chain, the second-tier supplier for battery cells is also considered. The battery cells are shipped from the second-tier supplier in Asia to the first-tier supplier in Germany with an average lead time of $l_{ij} = 20$.

Engine supply chain:

The engine supplier produces 27 different combustion engines and one electric engine. Its production is mainly assembly-driven, and the production lead time of edge $(\text{engine-supplier_prod}, \text{engine-supplier_inv})$ and the transportation lead time l_{ij} of edge $(i, j) = (\text{engine-supplier_inv}, \text{zp7})$ are 1.

Gear supply chain:

The gear supplier offers eight different gears for cars with conventional (e.g. ICE) and alternative (e.g. BEV) drive concepts. Due to the mechanically driven production, the production lead time l_{ij} of edge $(i, j) = (\text{gear-supplier_prod}, \text{gear-supplier_inv})$ and the transportation lead time l_{ij} of edge $(i, j) = (\text{gear-supplier_inv}, \text{zp7})$ are 2.

Seat supply chain:

Seats are hard to store efficiently and economically due to their wide variety and large volume. Thus, seats are BTO and are delivered just-in-time (JIT) at the time of assembly and delivered just-in-sequence (JIS) in accordance with the planned assembly order. This produces a transportation and production lead time of less than four hours. Thus, the lead times l_{ij} of edge $(i, j) = (\text{seat-supplier_prod}, \text{seat-supplier_inv})$ and $(i, j) = (\text{seat-supplier_inv}, \text{zp7})$, as well as inventory inv_{npt} of node $n = \text{zp7}$ for $g_p = \text{seat}$, are set to 0. Due to the importance of slow response delivery pro-

cesses for operational short-term decisions, the delivery process for seat components at edge $(i, j) = (\text{seatleather-supplier_out}, \text{battery-supplier_prod})$ with a lead time $l_{ij} = 23$ is also considered.

4.5. Explanation of model requirements

To fulfil the objectives of the planning task, the seven requirements below must be met:

- Period-related consideration of all stocks at all stages within the production network as well as planned production schedules
- Implementation of different production strategies (BTO, BTS)
- Setting of different production and delivery lead times
- Implementation of capacity restrictions for production and delivery strategies
- Implementation of converging networks
- BOM explosion
- Detection of sequence violations or supply shortages
- Consideration of the production schedules of firms within the production network

Parameters

γ	weighting factor of flow f_{ijpt}
δ	weighting factor of inventory inv_{npt}
ϵ	weighting factor of order shortfall os_{pt}
ζ	weighting factor of outflow of_{pt}
ζ_t	deviation factor based on scheduled period t and promised delivery date t_{np}^s at node $n = \text{zp8}$ of item p
η	weighting factor of unassigned capacities uc_t
η_t	deviation factor based on actual period t and latest period T

Decision variables

f_{ijpt}	flow of item p at edge (i, j) at period t
inv_{npt}	inventory of item p at node n at period t
os_{pt}	order shortfall at node $n = \text{zp8}$ of item p at period t
of_{pt}	outflow at node $n = \text{zp8}$ of item p at period t
uc_t	unassigned capacity at edge $(\text{zp7}, \text{zp8})$ at period t

4.6. Application of the model

In accordance with the application of generic models, the decision variables os_{pt} , of_{pt} and uc_t are added to the objective function to satisfy the requirements of the application domain (see target function (9)). The factors γ , δ , ϵ , ζ , ζ_t , η , and η_t are integrated to allow the weighting of each decision variable. To allow order shortfalls, the demand d_{npt} of constraint (4) of the generic optimisation model described in Section 3 is replaced by outflow of_{pt} .

$$\begin{aligned}
 \min \gamma & \sum_{i,j \in A} \sum_{p \in P} \sum_{t \in T} f_{ijpt} \\
 & + \delta \sum_{n \in N} \sum_{p \in P} \sum_{t \in T} inv_{npt} \\
 & + \epsilon \sum_{p \in P} \sum_{t \in T} os_{pt} \\
 & + \zeta \sum_{p \in P} \sum_{t \in T} (\zeta_t \cdot of_{pt}) \\
 & + \eta \sum_{t \in T} (\eta_t \cdot uc_t)
 \end{aligned} \quad (9)$$

To ensure a high level of customer service, order shortfalls and delivery deviations have to be minimised. Constraint (10) shows the calculation of order shortfall os_{pt} as the difference of the promised delivery date of the quoted demand d_{npt} at node $n = \text{zp8}$

and flow f_{ijpt} at edge $(i, j)=(zp7, zp8)$ for item $p \in P$. Delivery deviations are considered by multiplying the outflow of_{pt} of node $n=zp8$ with the deviation factor ζ_t of item $p \in P$ for period $t \in T$. As shown in constraint (11), the outflow of_{pt} equals the difference of demand d_{npt} and order shortfall os_{pt} at node $n=zp8$ of item p at period t . Based on the v-shaped value function in constraint (12), ζ_t sanctions the degree of deviation of actual scheduled period t from original scheduled period t_{np}^s . The left graph in Fig. 9 exemplifies different ζ_t for periods $\forall t \in T$ for $t_{np}^s = 67$. The optimal utilisation of the assembly line is an important key performance indicator for OEM production. Thus, the unassigned capacity uc_t is based on the difference of capacity c_{ijt} and the flow f_{ijpt} at edge $(i, j)=(zp7, zp8)$ of item $p \in P$ for period $t \in T$ (see constraint (14)). As shown in constraint (13) and illustrated in the right-hand graph of Fig. 9, the decreasing value function η_t ensures that the unassigned capacities of the current periods are sanctioned more highly than the unassigned capacities of subsequent periods.

$$os_p = d_{zp8,pt} - f_{zp7,zp8,pt} \quad \forall p \in P, \forall t \in T \quad (10)$$

$$\sum_{t \in T} of_{pt} = \sum_{t \in T} (d_{zp8,pt} - os_{pt}) \quad \forall p \in P \quad (11)$$

$$\zeta_t = \begin{cases} t_{np}^s - t, & t \leq t_{np}^s \\ t - t_{np}^s, & t > t_{np}^s \end{cases} \quad n = zp8, \forall p \in P, \forall t \in T \quad (12)$$

$$\eta_t = T - t + 1, \quad \forall t \in T \quad (13)$$

$$uc_t = c_{zp7,zp8,t} - \sum_{p \in P} f_{zp7,zp8,pt} \quad \forall t \in T \quad (14)$$

$$os_{pt}, of_{pt} \in \mathbb{N}_0 \quad \forall p \in P, \forall t \in T \quad (15)$$

$$uc_t \in \mathbb{N}_0 \quad \forall t \in T \quad (16)$$

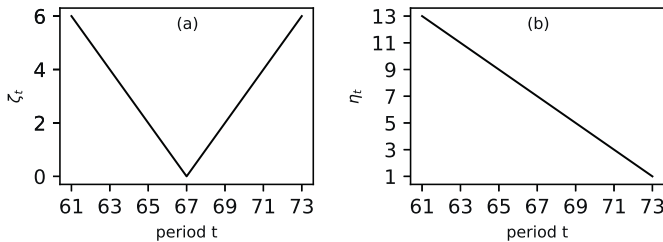


Fig. 9. Examples of applied value functions: (a) v-shaped value function ζ_t for $t_{np}^s = 67$, (b) decreasing value function η_t .

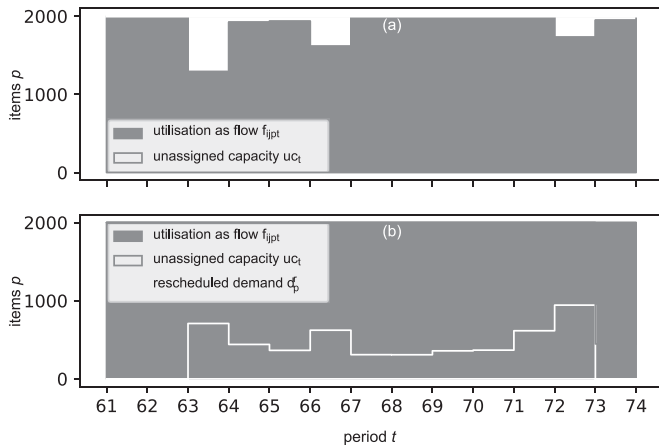


Fig. 10. Scenario comparison at edge $(i, j)=zp7, zp8$ and node $n=zp8$ for $g_p=car$: (a) scenario 1 without rescheduling with $\eta=0$ (b) scenario 2 with rescheduling with $\eta=200$.

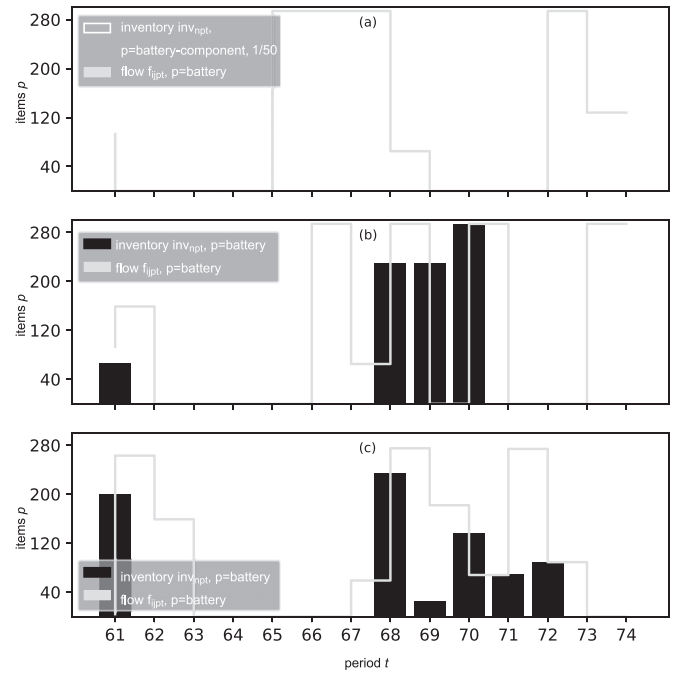


Fig. 11. Impact on battery supply chain: (a) $n=battery-supplier_prod$, $(i, j)=(battery-supplier_prod, battery-supplier_inv)$ (b) $n=battery-supplier_inv$, $(i, j)=(battery-supplier_inv, zp7)$ (c) $n=zp7$, $(i, j)=(zp7, zp8)$.

To consider the initial state of the production network, all initial inventories $inv_{np,ini}$ of p at node n are assigned to period $t=60$, whereas the initial flows f_{ijpt} of item p at edge (i, j) are assigned to periods $t < 60$ according to their initial start time. Furthermore, the observance of the supplier's scheduled production f_{ijpt} at edge (i, j) and item p for periods $t > 60$ is enforced via flow-related constraints. If we take the example of engine suppliers' scheduled production of period $t=65$ for item $p=D60$ with an amount of 40, the resulting constraint is shown in constraint (17).

$$f_{engine-supplier_prod, engine-supplier_inv, D60, 65} = 40 \quad (17)$$

4.7. Description of data source

The data source used corresponds to the random variation in real data, as no consistent or complete data on the initial inventories at all nodes or on the initial flows of all arcs are available (Rardin & Uzsoy, 2001). Thus, and as described in Table 8, the process to generate the data sets took place in 2017 and 2018 and encompassed four steps: data collection, data cleaning, model building, and model validation.

The random variations in practical data sets are realised by a simulation model that is based on system exports and a case study. The system exports are used to get a realistic distribution and structure of demand, as well as production network settings (for instance, lead times and capacities). Therefore, the OEM's demand is based on a real customer order pool. The case study data were obtained during an abductive case study conducted in 2017 (for details, see Moetz et al., 2018). Based on this process information and the insights obtained by the case study, an object-oriented model was built to generate consistent shortage scenarios. The validity of the model was ensured via empirical, structure, and decision validations. This included statistical design experiments to verify the structure and parameter values of the model (for instance, information processes and lead times) and its plausible results. Fur-

Table 8

Process steps to generate data sets.

Process steps	Description	Duration
Data Collection	Data sets are based on an explorative embedded case study within the production network of Volkswagen AG including 24 interviews with 22 informants as well as system exports (for detailed information on the case study, see Moetz et al. (2018))	5 Months for data collection
Data Cleaning	Detection of duplicates as well as merging and completion of incomplete data	2 months, parallel to the steps of data collection and model building
Model Building	Building the model	1 month
Model Validation	Empirical validation, structure validation and decision validation	3 months, parallel to the steps of data collection, data cleaning and model building

thermore, the model and scenarios were discussed with process experts at all stages of the network. By considering the transient response and a safety factor of 30%, the settling time was set to 55 periods. The complete data set used for this case study is available in [Moetz, Quetschlich, and Otto \(2020\)](#).⁹

4.8. Implementation

The model was built and tested using IBM ILOG CPLEX on a Windows 7 64 Bit system with an Intel i5-5300 U CPU and 16 GB RAM. The solution time was 53 seconds.

4.9. Results

The practical results of applying the optimisation model are twofold. First, the total impact of the supply shortage was evaluated; second, a rescheduling-driven reaction measure is proposed. The results are explained and compared in *scenario 1*: without rescheduling and *scenario 2*: with rescheduling, as shown in [Fig. 10](#). This comparison is based on the variables f_{ijpt} , unassigned capacity uc_{pt} , and rescheduled demand d_p^r at edge $(i, j) = (zp7, zp8)$ of item group $g_p = car$ for periods $62 \leq t \leq 67$.

Scenario 1: Without rescheduling

To prevent rescheduling in *scenario 1*, factors w are set as follows: $\gamma=1$, $\delta=1$, $\epsilon=3000$, $\zeta=100$, and $\eta=200$. Thus, order shortfalls and unassigned capacities are not sanctioned, whereas rescheduling is highly sanctioned. The top graph of [Fig. 10](#) shows the utilisation of the OEM's production through flow f_{ijpt} (grey shaded area) and unassigned capacity uc_{pt} (grey hatched area) for periods. In total, 1,552 BEV cannot be produced for periods $63 \leq t \leq 73$. The low utilisation is the consequence of the supply scarcity within the whole battery supply chain, as shown in [Fig. 11](#). The top graph shows node $n=battery-supplier_prod$, the central graph shows node $n=battery-supplier_inv$, and the lower graph shows node $n=zp7$ through their inventory inv_{npt} and the subsequent flow f_{ijpt} for $n=i$ of item group $g_p=battery-cell$ or $g_p=battery$ of periods $63 \leq t \leq 73$. The stock level at node $n=zp7$ drops from 200 batteries in period $t=61$ to 0 batteries for periods $62 \leq t \leq 67$ due to the non-existent flows of edge $(i, j) = (battery-supplier_inv, zp7)$ of periods $t \in (61, 63, 64, 65, 66)$ and of node $n=battery-supplier_prod$ for periods $62 \leq t \leq 65$. Through the receipt of new deliveries of battery cells in period $t=66$ at node $n=battery-supplier_prod$, full utilisation of the OEM's production is reached (indicated by the reduction of unassigned capacities shown in [Fig. 10](#)) for period $t \geq 68$, except for periods $t=72$ and $t=73$.

Scenario 2: With rescheduling

Rescheduling plans are generated by sanctioning order shortfalls and unassigned capacities in *scenario 2* ($\eta=0$ in this case). As the lower diagram in [Fig. 10](#) indicates, it is possible to equalise the unassigned capacities (black hatched area in the upper diagram)

of each period with pulled demand (white hatched space in the lower diagram) from subsequent periods. All affected customer orders (grey hatched area) are postponed to period $t=74$. Hence, as enabled by rescheduling, the OEM's production lines would attain optimal utilisation, and dispatchers would have more time to solve the delivery disruption. At the same time, the execution of the proposed production schedule is safeguarded due to the multi-echelon analysis with BOM explosion.

4.10. Discussion

First, this case study demonstrated how the generic framework can be customised to support short-term planning by evaluating the extent of supply shortages and by generating possible solving strategies. The simultaneous consideration of affected BOMs, material, inventories, and flows enables a comprehensive evaluation of the extent of shortages and facilitates the generation of executable reaction strategies. The latter is a particularly important contribution to the reduction of schedule instability in multi-echelon production networks. As explained, rescheduling methods and strategies do not consider the affected supply chains. Therefore, the supply reliability of altered production schedules cannot be guaranteed, which often results in iterative rescheduling cycles. To increase the adoption and utilisation of the model for cases like the planning task discussed above, two points should be considered in future research. First, while rescheduling measures have to consider the network-related supply of the altered production schedule, they must also ensure their plant-related realisation (e.g. technical constraints). Thus, network allocation (e.g. the presented model) and scheduling (e.g. car sequencing problems) models have to be integrated. Second, OEMs, suppliers, and third-party logistics providers use a variety of measures to cope with supply shortages. These measures have to be analysed and implemented, in order to enhance the information value of the generated scenarios.

5. Conclusions and future research

We would like to point out three key aspects of the model. First, the generic model is capable of combining any production, inventory, and transportation element as given in [Appendix Table 9](#). This enables the integrated planning of complex nested multi-echelon supply chains. Furthermore, the generic model can be used as a framework, whose building blocks can be 'plugged in' as modules. This modular framework contributes to the existent scientific research because most publications have so far focused on the integration of a few specific process steps and planning tasks and lack comprehensive integration. In the case study conducted with Volkswagen AG, we demonstrate that horizontal integration is possible with real industry data for seven entities: one assembly line, four first-tier suppliers, and two second-tier suppliers.

The second important aspect is the routing of multiple materials and transportation equipment with arbitrary BOM structures.

⁹ Also see <https://doi.org/10.17632/pr3sdy5vp3.1>

In addition to the planning of production periods, transportation periods, and inventory levels, the model facilitates the planning of cutting/separating and assembly processes. In the industry case study, 28 types of engine, eight types of gear, three types of battery, and multiple types of seat are assembled to form several different types of car.

The third aspect is that the generic model formulation can easily be extended. In the industry case study, we provide a specific model extension to accommodate order shortfalls and unassigned capacities caused by short-term supply shortages in a multi-echelon automotive production network of a German car manufacturer. Furthermore, the tasks explained in the supply chain planning matrix (Fig. 2) could be used as further extensions of the generic model.

Future research could take the following directions. First, to further generalise the model, the abovementioned extensions could be integrated into the model, as could additional rules and industry-specific constraints. Pinedo (2016) defines some specific rules, including *permutations*, *release dates*, *due dates*, *weighting*, *sequence dependent setup times*, *batch processing* or *no-wait*. In the automotive industry specific planning steps such as *line configura-*

tion, *line balancing* (see Boysen, Fliedner, & Scholl, 2008), *JIS*, *JIT*, or *Kitting*, could be integrated (see Hua & Johnson, 2010; Swaminathan & Nitsch, 2007).

Second, the model could be tested on more instances of real-world problems, similar to the case described in Section 4.

Third, the model could be tested to confirm that it can solve different types of integrated planning tasks with only minor adjustments. For the automotive industry, possible applications include *congestion management*, *capacity management* (Martínez-Costa, Mas-Machuca, Benedito, & Corominas, 2014), *strategic load and investment planning*, *distribution and outbound planning*, *reverse logistics*, (Boysen, Emde, Hoeck, & Kauderer, 2015; Govindan & Soleimani, 2017), *online ordering / order scheduling*, *late order assignment* (Meyr, 2004), *materials banking*, *planning the flow of materials in factories*, and the simulation of *special orders*.

Fourth, computational issues and solution strategies could be studied. These could include heuristics, decomposition strategies (as in Casazza, Ceselli, and Wolfier Calvo (2019), and new technologies.

Appendix A

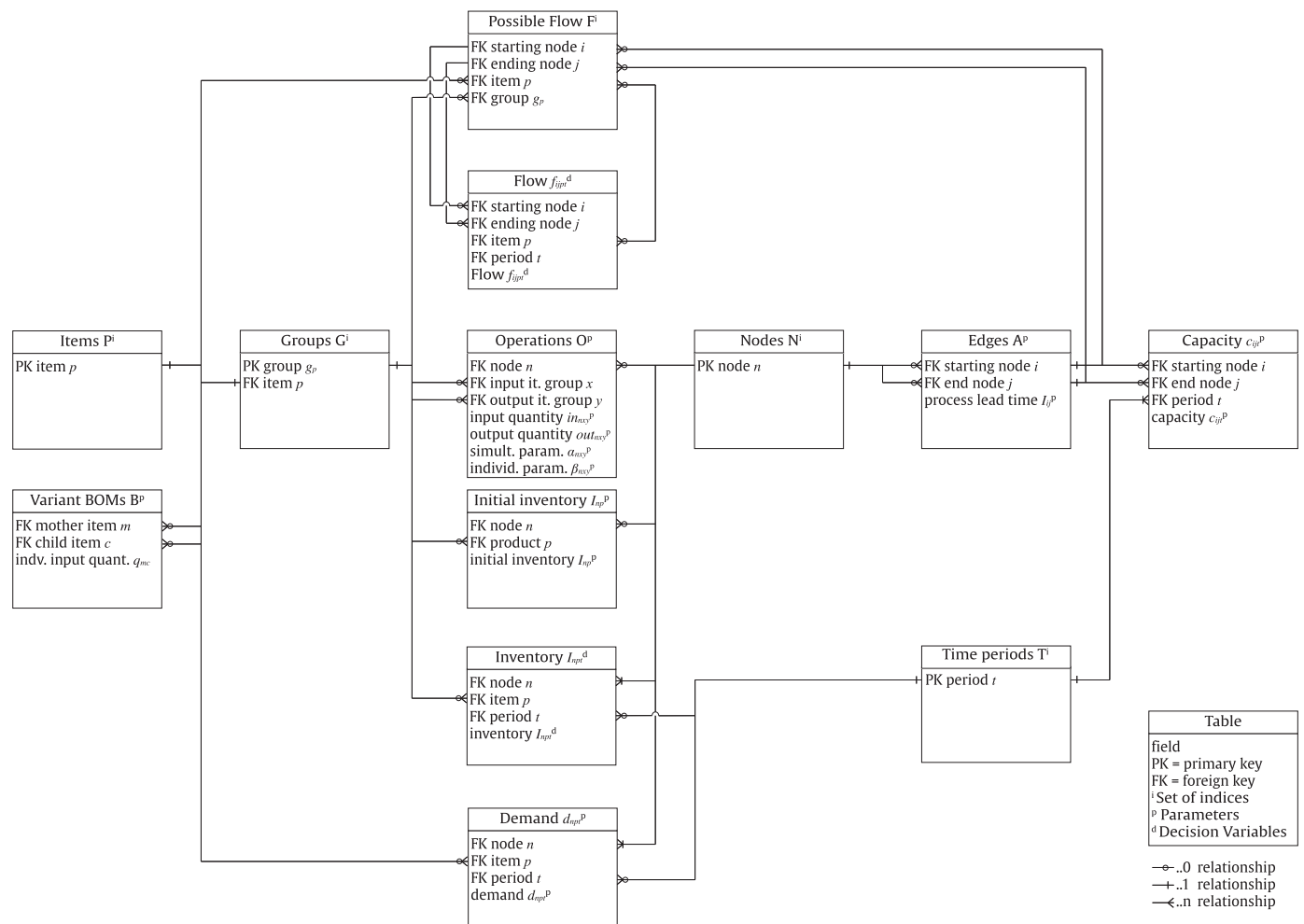
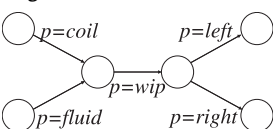
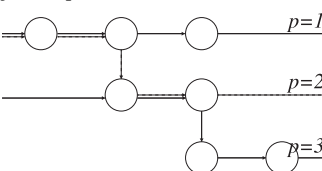
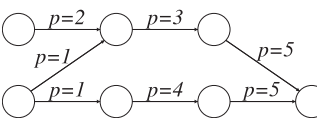
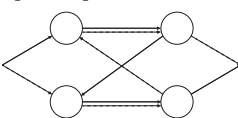
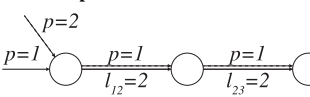
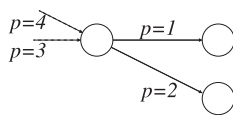
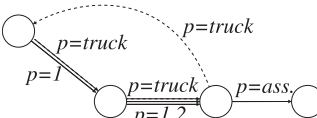
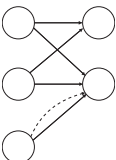
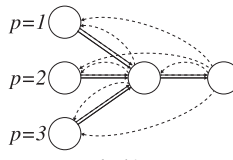


Fig. 12. Entity relationship model.

Table 9

Basic supply chain elements as building blocks.

Production	Single machine  All items p have to be processed on the same machine, represented by a single edge (i, j). On starting node i, a single machine can have one or more input item groups x, defined by one or more operations o. On ending node j, a single machine can have one or more output item groups y, also defined by one or more operations o. The machine's lead time is defined by l_{ij}. Job shop  In a job shop environment, items p are processed on two or more machines, represented by edges (i, j). The items can have different precedence relations, represented by different sets of operations. In this example, three different products are processed on different, partly overlapping machines.	Parallel machines  One or more final products p can be produced on two or more machines, which are modelled by multiple parallel edges (i, j) with process lead times l_{ij} and sets of operations o. In this example, two parallel machines produce the same final product $p = 5$. Open shop  Open shop problems generalise the job shop problem by neglecting some of the precedence relations: The item can visit multiple machines in any sequence. This results in two possible directed paths through the network. In the given example a product can be processed in any order on two machines. The two possible paths are represented by the dashed and the continuous line.	Flow shop  Flow shops can be modelled by a directed path, whereby each node n represents the starting node i of a workstation with lead time l_{ij}. The lead times can be estimated by the model as proposed by Jansen, Atan, Adan, and de Kok (2019). Edges (i, j) connect the workstation i with the following workstation j. If the assembly operations of specific materials must be assigned to specific workstations, the path can be extended as a converging directed graph. Continuous flow  A continuous flow is the production of equal products p on serially arranged process steps without inventory buffers inv_{npt}. In contrast to assembly lines, all finished products p have the same BOM. All process steps can be modelled as a directed graph with a single starting node $n = 1$ and one or more end nodes $n \neq 1$.
Inventory	Receiving and shipping Receiving and shipping consists of three problems: (1) Assignment of carriers to docks; (2) scheduling of carriers; (3) allocation of material handling resources (Gu, Goetschalckx, & McGinnis, 2007, p. 4). Docks can be modelled as nodes n. The schedule of carriers (p in $g = transport$) is given by flow f_{ijpt}. The allocation of material handling resources (3) can be modelled by transportation equipment p with $g = transport$.	Storage The task of storage answers (1) how much inventory should be kept; (2) how frequently and at what time the inventory should be replenished; (3) where the parts should be stored, distributed, and moved (Gu et al., 2007, p. 5). The first two questions are addressed by lot sizing problems, which are beyond the scope of this study (see Copil et al. (2017) and Glock, Grosse, and Ries (2014)). The third task can be modelled as a network of nodes n, edges a, and products p being stored. Milkrun  In milk run systems, transportation equipment (in $g = transport$) travel along a predefined cyclic route, loading and unloading items ($g \neq transport$). In the given example trucks travel along the path represented by the dashed lines, whereas products are routed through the network along the continuous lines.	Order picking According to Gu et al. (2007) order picking consists of the basic steps of <i>batching</i>, <i>routing</i>, <i>sequencing</i>, and <i>sorting</i>. Batching is not a focus of this model. Sorting is also not considered in this study. Both functions could be integrated into an extension of the model.
Transport	Point-to-point  In a point-to-point network, suppliers deliver products to their customers directly. A transport can be modelled by an edge (i, j) with the process lead time l_{ij}. Two nodes can be connected by multiple edges to model alternative transportation modes, such as expensive but fast air freight and cheaper but slower container ship transport.	Cross docking  In a cross-docking system, two or more input item groups g are unloaded, sorted, and consolidated in order to be shipped to the next node (e.g. to the OEM with batches that follow JIS demand). In this example three items are shipped towards the cross-docking terminal represented by node $n = 4$ and consolidated. The flow of items is represented by the continuous lines, and the flow of trucks by the dashed lines.	

References

- Absi, N., Archetti, C., Dauzère-Pérès, S., Feillet, D., & Speranza, M. G. (2018). Comparing sequential and integrated approaches for the production routing problem. *European Journal of Operational Research*, 269(2), 633–646.
- Arnold, D., Isermann, H., Kuhn, A., Tempelmeier, H., & Furmans, K. (2008). *Handbuch Logistik*. Berlin, Heidelberg: VDI-Buch, Springer.
- Arrow, K. J., Karlin, S., & Scarf, H. (1958). *Studies in the Mathematical Theory of Inventory and Production*. Stanford: Stanford University Press.
- Arshinder, K., Kanda, A., & Deshmukh, S. G. (2011). *Supply Chain Coordination under Uncertainty*. Berlin, Heidelberg: Springer.
- Atan, Z., Ahmadi, T., Stegehuis, C., de Kok, T., & Adan, I. (2017). Assemble-to-order systems: A review. *European Journal of Operational Research*, 261(3), 866–879.
- Bard, J. F., & Nananukul, N. (2010). A branch-and-price algorithm for an integrated production and inventory routing problem. *Computers & Operations Research*, 37(12), 2202–2217.
- Bellman, R. (1954). The theory of dynamic programming. *Bulletin of the American Mathematical Society*, 60(6), 503–516.
- Boysen, N., Emde, S., Hoeck, M., & Kauderer, M. (2015). Part logistics in the automotive industry: Decision problems, literature review and research agenda. *European Journal of Operational Research*, 242(1), 107–120.
- Boysen, N., Fliedner, M., & Scholl, A. (2008). Assembly line balancing: Which model to use when? *International Journal of Production Economics*, 111(2), 509–528.
- Boysen, N., Scholl, A., & Woppreper, N. (2012). Resequencing of mixed-model assembly lines: Survey and research agenda. *European Journal of Operational Research*, 216(3), 594–604.
- Calvete, H. I., Galé, C., & Polo, L. (2016). Integrated supply chain planning: A review. volume 254 of. *In Lecture notes in business information processing* (pp. 92–103). Cham: Springer International Publishing.
- Casazza, M., Ceselli, A., & Wolfler Calvo, R. (2019). A route decomposition approach for the single commodity split pickup and split delivery vehicle routing problem. *In Proceedings of the european journal of operational research*.
- Chen, Z. L. (2004). Integrated production and distribution operations. In D. Simchi-Levi, D. S. Wu, & Z. J. Shen (Eds.), *Handbook of quantitative supply chain analysis* (pp. 711–745). Boston, MA: Springer.
- Chen, Z. L. (2010). Integrated production and outbound distribution scheduling: Review and extensions. *Operations Research*, 58(1), 130–148.
- Choi, T. Y., Dooley, K. J., & Rungtusanatham, M. (2001). Supply networks and complex adaptive systems: control versus emergence. *Journal of Operations Management*, 19(3), 351–366.
- Clark, A. J. (1958). A dynamic, single-item, multi-echelon inventory model. (RM-2297).
- Clark, A. J., & Scarf, H. (1960). Optimal policies for a multi-echelon inventory problem. *Management Science*, 6(4), 475–490.
- Collette, Y., & Siarry, P. (2004). *Multiobjective Optimization*. Berlin, Heidelberg: Decision Engineering, Springer.
- Copil, K., Wörbelauer, M., Meyr, H., & Tempelmeier, H. (2017). Simultaneous lotsizing and scheduling problems: a classification and review of models. *OR Spectrum*, 39(1), 1–64.
- Díaz-Madroño, M., Peidro, D., & Mula, J. (2015). A review of tactical optimization models for integrated production and transport routing planning decisions. *Computers & Industrial Engineering*, 88, 518–535.
- Dörmer, J., Günther, H.-O., & Gujjula, R. (2015). Master production scheduling and sequencing at mixed-model assembly lines in the automotive industry. *Flexible Services and Manufacturing Journal*, 27(1), 1–29.
- Duncan, R. B. (1972). Characteristics of organizational environments and perceived environmental uncertainty. *Administrative Science Quarterly*, 17(3), 313.
- Ehm, J., & Freitag, M. (2016). The benefit of integrating production and transport scheduling. *Procedia CIRP*, 41, 585–590. (April)
- European Commission (2019). Internal market, industry, entrepreneurship and SMEs - automotive industry.
- Gansterer, M., & Hartl, R. F. (2018). Collaborative vehicle routing: A survey. *European Journal of Operational Research*, 268(1), 1–12.
- Gantt, H. L. (1918). Efficiency and democracy. *In Proceedings of the american society of mechanical engineers*, New York.
- Glock, C. H., Grosse, E. H., & Ries, J. M. (2014). The lot sizing problem: A tertiary study. *International Journal of Production Economics*, 155, 39–51.
- Goetschalckx, M., Vidal, C. J., & Dogan, K. (2002). Modeling and design of global logistics systems: A review of integrated strategic and tactical models and design algorithms. *European Journal of Operational Research*, 143(1), 1–18.
- Govindan, K., & Soleimani, H. (2017). A review of reverse logistics and closed-loop supply chains: a journal of cleaner production focus. *Journal of Cleaner Production*, 142(3), 371–384.
- Gu, J., Goetschalckx, M., & McGinnis, L. F. (2007). Research on warehouse operation: A comprehensive review. *European Journal of Operational Research*, 177(1), 1–21.
- Handelsblatt (2019). Streik bei audi in ungar beendete - einigung im tarifstreit.
- Harjankoski, I., Maravelias, C. T., Bongers, P., Castro, P. M., Engell, S., Grossmann, I. E., ... Wassick, J. (2014). Scope for industrial applications of production scheduling models and solution methods. *Computers and Chemical Engineering*, 62, 161–193.
- Harris, F. W. (1913). How many parts to make at once. *The Magazine of Management*, 10(2), 135–136.
- Hozak, K., & Hill, J. A. (2009). Issues and opportunities regarding replanning and rescheduling frequencies. *International Journal of Production Research*, 47(18), 4955–4970.
- Hua, S. Y., & Johnson, D. J. (2010). Research issues on factors influencing the choice of kitting versus line stocking. *International Journal of Production Research*, 48(3), 779–800.
- Jansen, S., Atan, Z., Adan, I., & de Kok, T. (2019). Setting optimal planned lead-times in configure-to-order assembly systems. *European Journal of Operational Research*, 273(2), 585–595.
- Johnson, S. M. (1954). Optimal two- and three-stage production schedules with setup times included. *Naval Research Logistics (NRL)*, 1(1), 61–68.
- Khoshnevis, B., & Chen, Q. M. (1991). Integration of process planning and scheduling— a review. *Journal of Intelligent Manufacturing*, 2(3), 165–175.
- Maleki, M., & Cruz-Machado, V. (2013). A review on supply chain integration: Vertical and functional perspective and integration models. *Economics and Management*, 18(2).
- Maravelias, C. T., & Sung, C. (2009). Integration of production planning and scheduling: Overview, challenges and opportunities. *Computers & Chemical Engineering*, 33(12), 1919–1930.
- Martel, A. (2005). The design of production-distribution networks: A mathematical programming approach. *In Supply chain optimization, number august 2004, pages 265–305*. New York: Springer-Verlag.
- Martínez-Costa, C., Mas-Machuca, M., Benedito, E., & Corominas, A. (2014). A review of mathematical programming models for strategic capacity planning in manufacturing. *International Journal of Production Economics*, 153, 66–85.
- Meyr, H. (2004). Supply chain planning in the german automotive industry. *OR Spectrum*, 26(4), 447–470.
- Meyr, H., & Stadler, H. (2015). Types of supply chains. In H. Stadler, C. Kilger, & H. Meyr (Eds.), *Supply chain management and advanced planning*, pages 55–69. Berlin, Heidelberg, 5th edition: Springer Texts in Business and Economics.
- Meyr, H., Wagner, M., & Rohde, J. (2008). Structure of advanced planning systems. *In Supply chain management and advanced planning* (pp. 109–115). Berlin, Heidelberg: Springer Berlin Heidelberg.
- Moetz, A., Meinschmidt, J., & Otto, B. (2018). Schedule instability in production networks: An empirical study on drivers and mechanisms within the automotive industry. *IFAC-PapersOnLine*, 51(11), 1071–1076.
- Moetz, A., Quetschlich, M., & Otto, B. (2020). Data for: Optimisation model for multi-item multi-echelon supply chains with nested multi-level products. *Mendeley Data*, v1.
- Moons, S., Ramaekers, K., Caris, A., & Arda, Y. (2017). Integrating production scheduling and vehicle routing decisions at the operational decision level: A review and discussion. *Computers & Industrial Engineering*, 104, 224–245.
- Mula, J., Peidro, D., Díaz-Madroño, M., & Vicens, E. (2010). Mathematical programming models for supply chain production and transport planning. *European Journal of Operational Research*, 204(3), 377–390.
- Paquet, M., Martel, A., & Desaulniers, G. (2004). Including technology selection decisions in manufacturing network design models. *International Journal of Computer Integrated Manufacturing*, 17(2), 117–125.
- Paquet, M., Martel, A., & Montreuil, B. (2008). A manufacturing network design model based on processor and worker capabilities. *International Journal of Production Research*, 46(7), 2009–2030.
- Pinedo, M. L. (2016). *Scheduling*. Cham: Springer International Publishing.
- Porter, M. E. (1985). *Competitive advantage*.
- Rardin, R. L., & Uzsoy, R. (2001). Experimental evaluation of heuristic optimization algorithms: A tutorial. *Journal of Heuristics*, 7(3), 261–304.
- Reimann, M., Tavares Neto, R., & Bogendorfer, E. (2014). Joint optimization of production planning and vehicle routing problems: A review of existing strategies. *Pesquisa Operacional*, 34(2), 189–214.
- Rizk, N., Martel, A., & D'Amours, S. (2006). Multi-item dynamic production-distribution planning in process industries with divergent finishing stages. *Computers and Operations Research*, 33(12), 3600–3623.
- Rizk, N., Martel, A., & Amours, S. D. (2004). Production planning in the pulp and paper industry production planning in the pulp and paper industry. *Technology*, 2004, 1–31.
- Rohde, J., Meyr, H., & Wagner, M. (2000). Die supply chain planning matrix. *PPS Management*, 5(2000), 10–15.
- Serdarasan, S. (2013). A review of supply chain complexity drivers. *Computers and Industrial Engineering*, 66(3), 533–540.
- Stadler, H., Kilger, C., & Meyr, H. (2015a). Supply chain management and advanced planning. *In Springer texts in business and economics*. Berlin, Heidelberg, 5th edition: Springer Berlin Heidelberg.
- Stadler, H., Kilger, C., & Meyr, H. (2015b). Supply chain management and advanced planning. *In Springer texts in business and economics*. Berlin, Heidelberg: Springer Berlin Heidelberg.
- Stonehouse, G., & Snowden, B. (2007). Competitive advantage revisited: Michael porter on strategy and competitiveness. *Journal of Management Inquiry*, 16(3), 256–273.
- Swaminathan, J. M., & Nitsch, T. R. (2007). Managing product variety in automobile assembly: The importance of the sequencing point. *Interfaces*, 37(4), 324–333.
- Timmer, M. P., Dietzenbacher, E., Los, B., Stehrer, R., & de Vries, G. J. (2015). An illustrated user guide to the world input-output database: the case of global automotive production. *Review of International Economics*, 23(3), 575–605.
- Vidal, C. J., & Goetschalckx, M. (1997). Strategic production-distribution models: A critical review with emphasis on global supply chain models. *European Journal of Operational Research*, 98(1), 1–18.